

# @modelcontextprotocol/server-filesystem

✓ SHIP-READY — all MUST checks pass, no security findings, no behavioural drift

|                   |                                                                                         |
|-------------------|-----------------------------------------------------------------------------------------|
| Server under test | npx -y @modelcontextprotocol/server-filesystem /tmp/mcp-proof-sandbox                   |
| Generated         | 2026-08-20 22:42 UTC · mcp-proof v0.6.0                                                 |
| Protocol          | negotiated 2025-11-25 · initialize-handshake era · newest initialize-handshake revision |
| Run fingerprint   | sha256: ff24d8392c55480748df4a255f66f8ba5a1664e01a71913568d3aed4de4da06a                |

CONFORMANCE

11/11

MUST checks passed · 7/7  
SHOULD

SECURITY

1

finding · 0 blocking · 1 advisory

BEHAVIOUR REGRESSION

34/34

replays clean · gate PASS

## Protocol conformance

| CHECK   | LEVEL  | RESULT | DETAILS                                                                                                                                            |
|---------|--------|--------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| LIFE-01 | MUST   | ✓ PASS | <b>initialize returns protocolVersion, capabilities and serverInfo</b><br>protocolVersion, capabilities and serverInfo all present                 |
| LIFE-02 | SHOULD | ✓ PASS | <b>server negotiates the newest handshake revision (2025-11-25)</b><br>negotiated 2025-11-25, the newest revision the initialize handshake carries |
| LIFE-03 | MUST   | ✓ PASS | <b>server answers tools/list after the initialized notification</b><br>tools/list returned 14 tool(s) after initialized                            |
| LIST-01 | MUST   | ✓ PASS | <b>tools/list pagination terminates (no cursor loop)</b><br>single page, no pagination cursor                                                      |
| RPC-01  | MUST   | ✓ PASS | <b>unknown method gets a JSON-RPC error response</b><br>unknown method rejected with error code -32601                                             |
| RPC-02  | SHOULD | ✓ PASS | <b>unknown method error code is -32601 (method not found)</b><br>expected -32601, got -32601                                                       |
| RPC-03  | MUST   | ✓ PASS | <b>malformed tools/call params are rejected with an error</b><br>malformed params rejected with error code -32603                                  |

| CHECK     | LEVEL  | RESULT | DETAILS                                                                                                                                                                                          |
|-----------|--------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T00L-01   | MUST   | ✓ PASS | <b>every tool has a non-empty name and an inputSchema</b><br>all 14 tools have a name and an inputSchema                                                                                         |
| T00L-02   | SHOULD | ✓ PASS | <b>every tool has a non-empty description</b><br>all 14 tools carry a description                                                                                                                |
| T00L-03   | MUST   | ✓ PASS | <b>every tool inputSchema compiles as JSON Schema</b><br>all 14 inputSchemas compile as JSON Schema draft 2020-12                                                                                |
| T00L-04   | MUST   | ✓ PASS | <b>calling a nonexistent tool is rejected</b><br>rejected as tool result with isError=true                                                                                                       |
| T00L-05   | MUST   | ✓ PASS | <b>a call missing required arguments is rejected</b><br>read_file with empty args rejected via isError=true                                                                                      |
| T00L-06   | MUST   | ✓ PASS | <b>declared outputSchemas compile and structured results match them</b><br>all 14 outputSchema(s) compile; dynamic call to read_file did not yield a normal result, structured output unverified |
| T00L-07   | SHOULD | ✓ PASS | <b>declared input constraints are enforced</b><br>6 schema-violating input(s) across 3 tool(s), all rejected                                                                                     |
| RES-01    | MUST   | – SKIP | <b>advertised resources capability serves resources/list</b><br>resources capability not advertised                                                                                              |
| RES-02    | MUST   | – SKIP | <b>every resource carries a uri and a name</b><br>resources/list unavailable                                                                                                                     |
| RES-03    | MUST   | – SKIP | <b>resources/read returns contents for an advertised resource</b><br>no listed resource to read                                                                                                  |
| RES-04    | MUST   | – SKIP | <b>resources/list pagination terminates (no cursor loop)</b><br>no surface to paginate                                                                                                           |
| PROMPT-01 | MUST   | – SKIP | <b>advertised prompts capability serves prompts/list</b><br>prompts capability not advertised                                                                                                    |
| PROMPT-02 | MUST   | – SKIP | <b>every prompt has a name and well-formed argument metadata</b><br>prompts/list unavailable                                                                                                     |
| PROMPT-03 | MUST   | – SKIP | <b>prompts/get rejects a call missing required arguments</b><br>no prompt declares required arguments                                                                                            |
| CAP-02    | SHOULD | ✓ PASS | <b>declared capabilities match served features (resources)</b><br>capabilities.resources not declared and resources/list not served                                                              |

| CHECK  | LEVEL  | RESULT | DETAILS                                                                                                                       |
|--------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------|
| CAP-03 | SHOULD | ✓ PASS | <b>declared capabilities match served features (prompts)</b><br>capabilities.prompts not declared and prompts/list not served |
| HYG-01 | MUST   | ✓ PASS | <b>stdout carries only JSON-RPC messages</b><br>no non-JSON-RPC stdout lines observed                                         |
| CAP-01 | SHOULD | ✓ PASS | <b>declared capabilities match served features (tools)</b><br>capabilities.tools declared and tools/list served               |

## Security & hygiene

| CHECK  | DOMAIN                  | RESULT | DETAILS                                                                                                                                                                                                                                                                                                                                                                                                                      |
|--------|-------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEC-01 | MCP-INPUT-01            | ✓ PASS | <b>no prompt-injection patterns in tool descriptions</b><br>0 matches across 14 tools                                                                                                                                                                                                                                                                                                                                        |
| SEC-02 | MCP-INPUT-01            | ✓ PASS | <b>no invisible or bidi control characters in tool metadata</b><br>0 invisible characters across 14 tools                                                                                                                                                                                                                                                                                                                    |
| SEC-03 | MCP-LOG-02              | ✓ PASS | <b>no secret-looking strings in tool metadata</b><br>0 secret-like strings across 14 tools                                                                                                                                                                                                                                                                                                                                   |
| SEC-04 | MCP-INPUT-02            | ! WARN | <b>injection-surface string params carry constraints</b><br>unconstrained injection-surface params: read_file.path, read_text_file.path, read_media_file.path, write_file.path, edit_file.path, create_directory.path, list_directory.path, list_directory_with_sizes.path, directory_tree.path, search_files.path, get_file_info.path<br><b>Fix:</b> Add enum, pattern or maxLength to path/url/command-like string params. |
| SEC-05 | MCP-INPUT-01            | ✓ PASS | <b>tool descriptions stay under 2000 chars</b><br>longest description 457 chars across 14 tools                                                                                                                                                                                                                                                                                                                              |
| SEC-06 | MCP-EXEC-01,MCP-EXEC-02 | ✓ PASS | <b>no tool advertises unconstrained arbitrary execution</b><br>0 exec-style tools with free-form params across 14 tools                                                                                                                                                                                                                                                                                                      |

## MSSS compliance

**L1: 2/2 auto-assessable controls met · 4 require manual review.** Mapped against MSSS v0.1 (control-level mapping v2.0 (2026-01-20)): 3 of 24 controls are auto-assessable from this audit's deterministic checks; the remaining 21 need deployment, code or process evidence and are marked manual review — never assessed by this tool.

| CONTROL       | DOMAIN           | LEVEL | RESULT          | TITLE                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|------------------|-------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| MCP-EXEC-01   | Execution        | L1    | – manual review | <b>Prohibition of Shell Execution</b><br>SEC-06 flags advertised arbitrary-exec tools, but proving the absence of shell invocation requires source review.                                                                                                                                                                                                                                                                                                                   |
| MCP-FS-01     | Filesystem       | L1    | – manual review | <b>Path Allowlisting and Canonical Resolution</b>                                                                                                                                                                                                                                                                                                                                                                                                                            |
| MCP-FS-02     | Filesystem       | L1    | – manual review | <b>Symlink Resolution Validation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| MCP-NET-01    | Network          | L1    | – manual review | <b>URL Validation and SSRF Mitigation</b>                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| MCP-INPUT-01  | Input Validation | L1    | ✓ met           | <b>JSON Schema Validation</b><br><b>Evidence checks:</b> TOOL-01, TOOL-03, RPC-03, TOOL-05, SEC-01, SEC-02, SEC-05<br>Auto-assessed from advertised schemas (present and valid), live rejection probes, and the tool-metadata poisoning scan.                                                                                                                                                                                                                                |
| MCP-LOG-02    | Logging          | L1    | ✓ met           | <b>Secret Redaction in Logs</b><br><b>Evidence checks:</b> SEC-03<br>Auto-assessed over advertised tool metadata; server log output is not inspected.                                                                                                                                                                                                                                                                                                                        |
| MCP-SUPPLY-02 | Supply Chain     | L2    | – manual review | <b>Trusted Package Sources</b>                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| MCP-INPUT-02  | Input Validation | L2    | ✓ met           | <b>Input Bounds Enforcement</b><br><b>Evidence checks:</b> SEC-04<br>SEC-04: unconstrained injection-surface params: read_file.path, read_text_file.path, read_media_file.path, write_file.path, edit_file.path, create_directory.path, list_directory.path, list_directory_with_sizes.path, directory_tree.path, search_files.path, get_file_info.path<br>Auto-assessed from advertised schema constraints (enum/pattern/maxLength); runtime payload limits are not probed. |
| MCP-INPUT-03  | Input Validation | L2    | – manual review | <b>Timeout Enforcement</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| MCP-NET-03    | Network          | L2    | – manual review | <b>TLS Enforcement</b><br>MSSS marks this N/A for stdio-only deployments with no network access.                                                                                                                                                                                                                                                                                                                                                                             |
|               |                  |       |                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

| CONTROL       | DOMAIN        | LEVEL | RESULT          | TITLE                                                                                                                               |
|---------------|---------------|-------|-----------------|-------------------------------------------------------------------------------------------------------------------------------------|
| MCP-EXEC-02   | Execution     | L2    | – manual review | <b>Command Allowlisting</b><br>See the SEC-06 advisory for advertised command tools; allowlist verification requires source review. |
| MCP-EXEC-03   | Execution     | L2    | – manual review | <b>Argument Separator Usage</b>                                                                                                     |
| MCP-AUTHZ-01  | Authorization | L3    | – manual review | <b>OAuth Token Delegation</b><br>MSSS marks this N/A for stdio transport with OS-level user isolation.                              |
| MCP-AUTHZ-02  | Authorization | L3    | – manual review | <b>Per-Tool Scope Definition</b>                                                                                                    |
| MCP-AUTHZ-03  | Authorization | L3    | – manual review | <b>Least Privilege Tool Design</b>                                                                                                  |
| MCP-AUTHZ-04  | Authorization | L3    | – manual review | <b>Resource-Based Access Control</b>                                                                                                |
| MCP-LOG-01    | Logging       | L3    | – manual review | <b>Comprehensive Audit Logging</b>                                                                                                  |
| MCP-DEPLOY-01 | Deployment    | L3    | – manual review | <b>Container Hardening</b>                                                                                                          |
| MCP-FS-03     | Filesystem    | L4    | – manual review | <b>Filesystem Sandboxing</b>                                                                                                        |
| MCP-SUPPLY-01 | Supply Chain  | L4    | – manual review | <b>Package Integrity Verification</b>                                                                                               |
| MCP-DEPLOY-03 | Deployment    | L4    | – manual review | <b>Resource Limits and Rate Limiting</b>                                                                                            |
| MCP-NET-02    | Network       | L4    | – manual review | <b>Egress Traffic Filtering</b>                                                                                                     |
| MCP-DEPLOY-02 | Deployment    | L4    | – manual review | <b>System Call Filtering (seccomp/AppArmor)</b>                                                                                     |
| MCP-DEPLOY-04 | Deployment    | L4    | – manual review | <b>Runtime Integrity Monitoring</b><br>Future control in MSSS v0.1 — implementation details TBD upstream.                           |

Control taxonomy from the MCP Server Security Standard (MSSS), CC BY-SA 4.0, mcp-security-standard.org.

### Behaviour regression

| FIXTURE                  | TOOL      | VERDICT | DETAIL |
|--------------------------|-----------|---------|--------|
| read_file__e8cf9eda.json | read_file | OK      |        |
|                          |           |         |        |

| FIXTURE                                  | TOOL                      | VERDICT | DETAIL |
|------------------------------------------|---------------------------|---------|--------|
| read_file__0c5f5c79.json                 | read_file                 | OK      |        |
| read_file__fe0dcdc6.json                 | read_file                 | OK      |        |
| read_file__518341a7.json                 | read_file                 | OK      |        |
| read_text_file__e8cf9eda.json            | read_text_file            | OK      |        |
| read_text_file__0c5f5c79.json            | read_text_file            | OK      |        |
| read_text_file__fe0dcdc6.json            | read_text_file            | OK      |        |
| read_text_file__518341a7.json            | read_text_file            | OK      |        |
| read_media_file__e8cf9eda.json           | read_media_file           | OK      |        |
| read_media_file__0c5f5c79.json           | read_media_file           | OK      |        |
| read_media_file__fe0dcdc6.json           | read_media_file           | OK      |        |
| read_media_file__518341a7.json           | read_media_file           | OK      |        |
| read_multiple_files__dfea4edf.json       | read_multiple_files       | OK      |        |
| list_directory__e8cf9eda.json            | list_directory            | OK      |        |
| list_directory__0c5f5c79.json            | list_directory            | OK      |        |
| list_directory__fe0dcdc6.json            | list_directory            | OK      |        |
| list_directory__518341a7.json            | list_directory            | OK      |        |
| list_directory_with_sizes__e8cf9eda.json | list_directory_with_sizes | OK      |        |
| list_directory_with_sizes__0c5f5c79.json | list_directory_with_sizes | OK      |        |
| list_directory_with_sizes__fe0dcdc6.json | list_directory_with_sizes | OK      |        |
| list_directory_with_sizes__518341a7.json | list_directory_with_sizes | OK      |        |
| directory_tree__e8cf9eda.json            | directory_tree            | OK      |        |
|                                          |                           |         |        |

| FIXTURE                                 | TOOL                     | VERDICT | DETAIL |
|-----------------------------------------|--------------------------|---------|--------|
| directory_tree__0c5f5c79.json           | directory_tree           | OK      |        |
| directory_tree__fe0dc6dc6.json          | directory_tree           | OK      |        |
| directory_tree__518341a7.json           | directory_tree           | OK      |        |
| search_files__580853a9.json             | search_files             | OK      |        |
| search_files__83e4c665.json             | search_files             | OK      |        |
| search_files__2831e0f1.json             | search_files             | OK      |        |
| search_files__ee546753.json             | search_files             | OK      |        |
| get_file_info__e8cf9eda.json            | get_file_info            | OK      |        |
| get_file_info__0c5f5c79.json            | get_file_info            | OK      |        |
| get_file_info__fe0dc6dc6.json           | get_file_info            | OK      |        |
| get_file_info__518341a7.json            | get_file_info            | OK      |        |
| list_allowed_directories__44136fa3.json | list_allowed_directories | OK      |        |

## Leave-behind: your regression gate

The recorded fixtures (demo/fixtures–filesystem, suite fingerprint sha256:866dc865fd9ca3735355ca4afe2b070c40fc8ce8d6fd072ae4bcf34e94803bbb) are the behavioural contract of this server. Keep them in the repository and run the workflow below in CI — any breaking or value drift fails the build before it reaches your users.

```
# The recorded fixtures in demo/fixtures–filesystem are the behavioral contract: any drift fails this
name: mcp-proof regression gate
on:
  push:
  pull_request:
jobs:
  replay:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - uses: actions/setup-python@v5
      with:
```

```
python-version: "3.11"
- name: Install mcp-proof
  run: pip install git+https://github.com/YuCPbit/mcp-proof
- name: Replay golden fixtures against the live server
  run: mcp-proof replay --fixtures demo/fixtures-filesystem -- npx -y @modelcontextprotocol/se
```

## Recommended next steps

**P1** SEC-04 — Add enum, pattern or maxLength to path/url/command-like string params.

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Reproducible audit — the fingerprint covers every check outcome and behavioural replay verdict (never timestamps or latency measurements). Re-run with identical server behaviour and the hash matches: sha256:ff24d8392c55480748df4a255f66f8ba5a1664e01a71913568d3aed4de4da06a

Generated by mcp-proof v0.6.0 · deterministic lanes only, no LLM judging · MCP spec reference 2025-11-25 (initialize-handshake era)